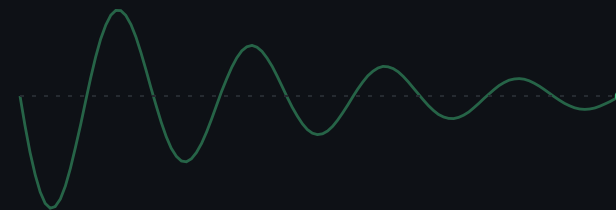


# zicato

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a self-improving meta-harness for multi-agent systems –  
evolution under a **protected incumbent**.

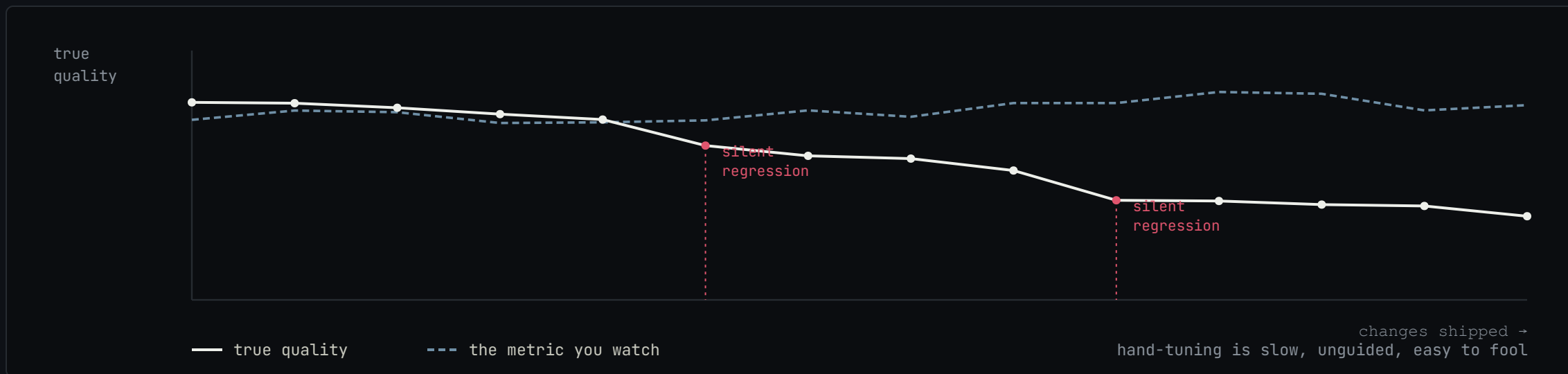
a sibling of goldfive + harmonograf



loss settling, generation over generation

# Agentic systems drift.

As prompts, tools, models and tasks change, quality silently regresses – and you usually can't tell whether a change helped or hurt.



# Measuring real improvement without ground truth.

Four pressures fight every attempt to tell a good change from a bad one.

01

## NOISY

LLM stochasticity.

One duel misleads – the same pair re-run disagrees.  
A single win can be luck.

02

## EXPENSIVE

A board run costs budget.

Every evaluation burns tokens and wall-clock. You cannot re-measure for free.

03

## A PROXY

Goodhart's law.

The metric is a stand-in. Push it hard and you game it – you overfit the board.

04

## NEVER REGRESS

Protect what worked.

A change must never silently break a task that passed before. Asymmetric risk.

The crux: get a signal that is real, cheap enough, hard to game, and regression-guarded.

# A champion / challenger loop.

Mutate the harness into a challenger, race it against the reigning champion, promote on a gate.



## NESTED COUNTERS

### epochs

one frozen evaluation contract

### rounds

one propose / duel / gate

### generations

the field a round mints

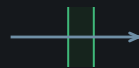
generations nest in rounds nest in epochs

# zicato composes proven selection theory.

Each borrowed idea is a small, well-understood mechanism. The loop is their composition.

## GATING / MONOTONIC IMPROVEMENT

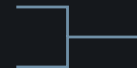
evaluator match + trust regions



AlphaZero-style: promote only on a beat; RL never steps past a region gate

## TOURNAMENTS & SUCCESSIVE HALVING

Swiss / elimination / racing



Hyperband from AutoML & bandits – spend evaluation where it counts  
race

## REPLICATION = THE NOISE LEVER

best-arm identification



re-run the duel to shrink variance;  
more samples, not more brackets  
replicate

## INCUMBENT PROTECTION = ELITISM

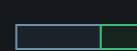
evolution strategies



the champion always survives unless genuinely beaten – elitist carry elitism

## ADAPTIVE-DATA-ANALYSIS DEFENCES

the reusable holdout / the Ladder



Dwork et al. & Blum-Hardt: query a fixed set without using it up  
holdout

## zicato

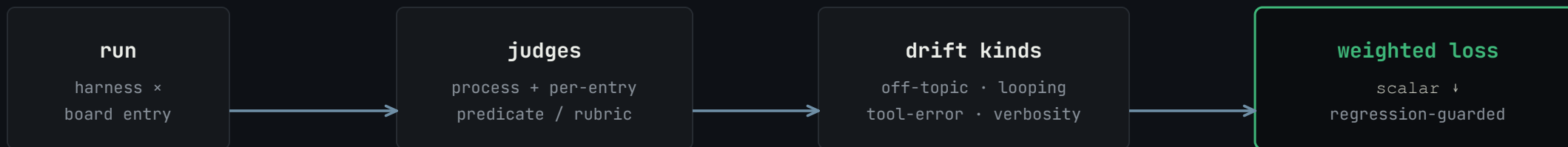
the composition



gate + tournament + replication + elitism + holdout, made to bite on agent behaviour

# goldfive's judges turn behaviour into a shaped loss.

Most loops have only coarse pass / fail. zicato runs the inner harness on goldfive – whose custom judges measure **drift**.



## THE WEIGHTED SCALAR

```
scalar = drift_weight · drift_loss + pass_weight · (1 - pass_rate)
```

|                          |                                |
|--------------------------|--------------------------------|
| <b>drift_weight</b>      | weight on cleanness            |
| <b>pass_weight</b>       | weight on correctness          |
| <b>per_kind_weights</b>  | elevate / demote a drift kind  |
| <b>per_judge_weights</b> | one step finer – by judge_name |

## YOU DEFINE WHAT “GOOD” MEANS

in-run process judges + per-entry predicates and rubrics fold into one rich, shaped, regression-guarded signal – where other loops have only thumbs up/down.

**This signal is what makes the loop work.**

# Promotion needs three things at once.

On reject, the champion stands. Asymmetric by design – the loop reduces drift without breaking what worked.

## 1 scalar margin

loss drops by  $\geq \text{promote\_margin}$

## 2 pass-rate monotonicity

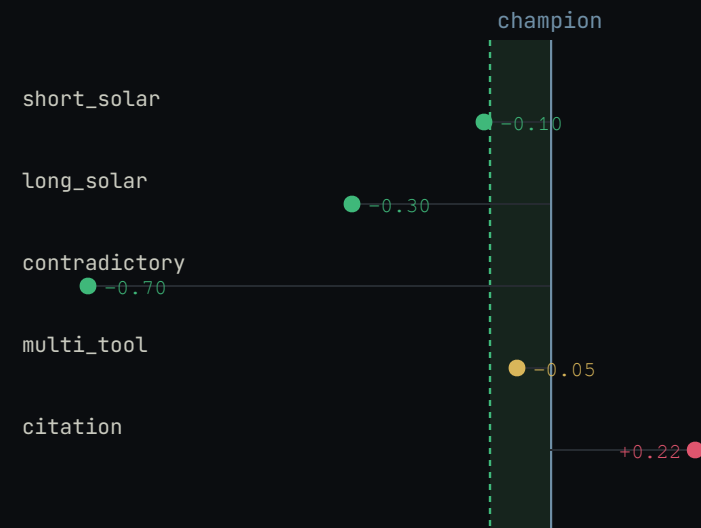
no previously-passing entry regresses

## 3 no per-task regression

guarded namespaces may not worsen

promote  $\Leftrightarrow$  all three hold. else: reject, champion stands.

$\Delta$  vs CHAMPION · per-entry loss



← improvement      promote\_margin      regression →  
green clears the margin · amber too small · red regresses → reject

# Configurable per epoch – but the gate is invariant.

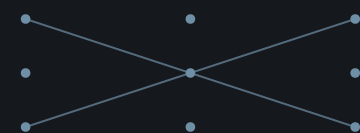
Replication, not bracket shape, is the noise lever. The gate, not the bracket, protects the incumbent.

default



**gauntlet**

default · 1v1 King-of-the-hill



**swiss**

Copeland ranking



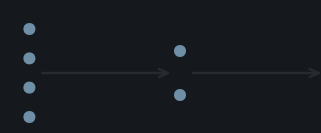
**single\_elim**

knockout bracket



**double\_elim**

a second life



**racing**

successive halving

## RACING – successive halving (the one bracket zicato endorses)



rung 0  
% board



rung 1  
% board



rung 2  
% board



final  
full board · GATE

eliminate the worst by scalar each rung · the GATE fires only at the full-board final



# The proposer is part of the evaluation contract.

Each round it reads the brief, mutation manifest, and loss patterns, and emits one `Experiment`.

## (a) SKILL-COMPOSED DEFAULT

drop in skills/\*.md – no code

- ☐ grounding.md
- ☐ house-style.md
- ☐ checklist.md

injected into the system prompt;  
steers HOW the default proposer reasons.  
runs on the auxiliary model.

## (b) CUSTOM ADK AGENT

drop in agent.py – its own model

agent.py

tool-using • own model

- read mutation
- propose patch
- reason

its presence selects the tier-(b) path;  
its contents are part of the contract.  
you own the model + the tools.

## CONTRACT

Changing the proposer or its skills `rolls a fresh epoch` – just like editing the board.  
generations proposed under different proposers are not directly comparable.

# Overfitting defences – default-on.

The proposer adaptively queries a fixed board, so it can **memorize** it (Goodhart / adaptive data analysis).

## TRAIN / HOLDOUT SPLIT

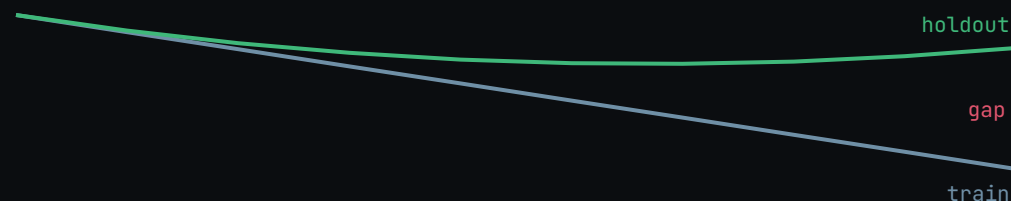


train slice

holdout

proposer + detectors see train only; promotion confirmed on holdout.

## GENERALIZATION GAP



gap widens → fake progress → end the epoch

### holdout-gated promotion

Ladder / Thresholdout,  
budgeted

### generalization-gap detector

a zicato health  
check

### holdout rotation

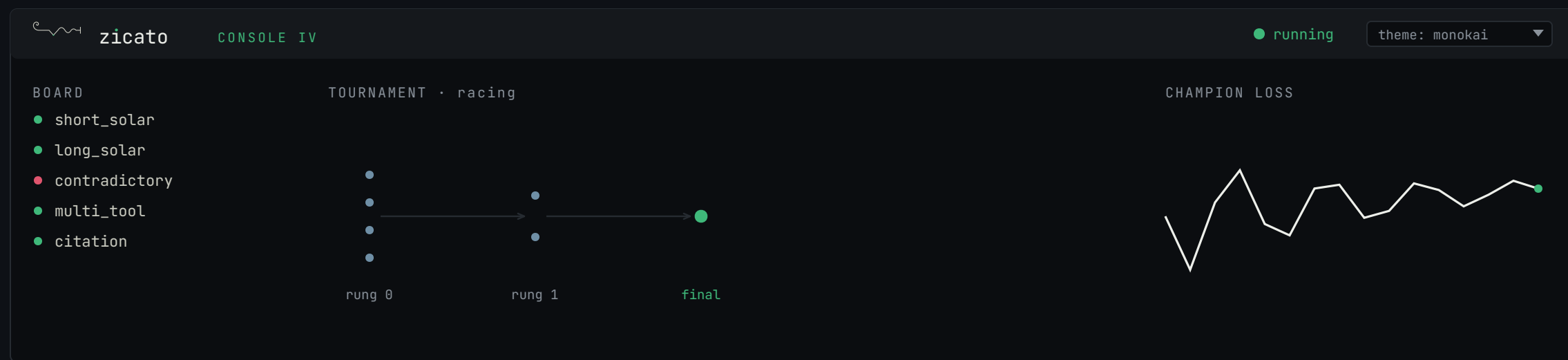
no slice mined  
forever

### proposer leakage restriction

patterns aggregated, inputs  
withheld

# Console IV – a dense observatory.

Live per-structure figures and board status, over a Rust watchdog and JSONL telemetry.



## 14 themes

monokai default + 13 Gogh palettes, swatch dropdown

## tournament builder

a chat copilot edits the draft contract

## configurable models

endpoints per harness / auxiliary / proposer



The thesis, in one line.

**Known-good selection theory, made to bite on agent behaviour  
via goldfive's drift judges –**  
promote only what genuinely, durably beats the incumbent.

WHAT'S NEXT

- Ranked Pairs
  - Bradley-Terry
  - maximal lotteries
- replicate-first, resolve-second.

zicato